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Semiconductor device and method of fabricating a semiconductor device

Abstract

A semiconductor device including an embedded channel structure, a sidewall channel structure and a gate electrode structure is provided. The embedded channel structure is disposed on a substrate. The sidewall channel structure is disposed on the substrate, and located at a lateral side of the embedded channel structure. The gate electrode structure is disposed on the substrate, encircles the embedded channel structure and is located between the embedded channel structure and the sidewall channel structure.

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Background/Summary

BACKGROUND

Technical Field

(1) The disclosure is related to a semiconductor device and a method of fabricating a semiconductor device.

Description of Related Art

(2) Semiconductor devices are used in a variety of electronic applications, such as personal computers, cell phones, digital cameras, and other electronic equipment. Semiconductor devices are typically fabricated by sequentially depositing insulating or dielectric layers, conductive layers, and semiconductor layers of material over a semiconductor substrate, and patterning the various material layers using lithography and etching techniques to form circuit components and elements thereon.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The accompanying drawings are included to provide a further understanding of the disclosure, and are incorporated in and constitute a part of this specification. The drawings illustrate exemplary embodiments of the disclosure and, together with the description, serve to explain the principles of the disclosure.
- (2) FIG. 1 to FIG. 6 schematically illustrate a method of fabricating a semiconductor device from a cross sectional view in accordance with some embodiments of the disclosure.
- (3) FIG. 7 to FIG. 12 schematically illustrate a method of fabricating a semiconductor device from a plane view in accordance with some embodiments of the disclosure.
- (4) FIG. 13 to FIG. 20 schematically illustrate a method of fabricating a semiconductor device from a cross section in accordance with some embodiments of the disclosure.
- (5) FIG. 21 to FIG. 27 schematically illustrate a method of fabricating a semiconductor device from a cross section in accordance with some embodiments of the disclosure.
- (6) FIG. 28 to FIG. 35 schematically illustrate a method of fabricating a semiconductor device from a cross section in accordance with some embodiments of the disclosure.

DESCRIPTION OF THE EMBODIMENTS

- (7) The following disclosure provides many different embodiments, or examples, for implementing different features of the invention. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. Unless other specified, the same or similar reference numerals in different figures refer to the same or similar component formed by a same or similar process(es) using a same or similar material(s).
- (8) Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.
- (9) The gate all around (GAA) transistor structures may be patterned by any suitable method. For example, the structures may be patterned using one or more photolithography processes, including double-patterning or multi-patterning processes. Generally, double-patterning or multi-patterning processes combine photolithography and self-aligned processes, allowing patterns to be created that have, for example, pitches smaller than what is otherwise obtainable using a single, direct photolithography process. For example, in one embodiment, a sacrificial layer is formed over a substrate and patterned using a photolithography process. Spacers are formed alongside the patterned sacrificial layer using a self-aligned process. The sacrificial layer is then removed, and the remaining spacers may then be used to pattern the GAA structure.
- (10) In some embodiments, a gate-all-around (GAA) field-effect transistor (FET) device includes a semiconductor strip protruding above a substrate, and a first isolation region and a second isolation region on opposing sides of the semiconductor strip. The GAA FET device also includes nanowires over and aligned with the semiconductor strip, and source/drain regions at opposing ends of the

nanowires. The GAA FET device further includes a first dielectric fin on the first isolation region, and a gate electrode around the nanowires and around center portions of the first dielectric fin, where end portions of the first dielectric fin are disposed on opposing sides of the gate electrode, and the end portions of the first dielectric fin are wider than the center portions of the first dielectric fin.

(11) FIG. 1 to FIG. 6 schematically illustrate a method of fabricating a semiconductor device from a cross sectional view in accordance with some embodiments of the disclosure. Referring to the X-axis, the Y-axis and the Z-axis, FIG. 1 to FIG. 6 depict the cross section of the structure cut along Y-axis under respective steps. In FIG. 1, a substrate **110** including a semiconductor strip **112** and an isolation structure **114** is provided. The semiconductor strip **112** is a protruded structure of the substrate **110** and is laterally surrounded by the isolation structure **114**. An epitaxial stack **120** may be disposed on the semiconductor strip **112** and specifically, on top of the semiconductor strip **112**. In addition, an epitaxial cap **130** is formed on the epitaxial stack **120** and on the isolation structure **114**. The epitaxial cap **130** may cover the top surface and the sidewall of the epitaxial stack **120** to form a cap-like shape. In some embodiments, the substrate **110** may have a plurality of semiconductor strips **112** each extending along the X-axis, and FIG. 1 presents the structure that one semiconductor strip **112** is cut along the width direction, the Y-axis, of the semiconductor strip **112** for illustration purpose.

(12) The substrate **110** may be a semiconductor substrate, such as a bulk semiconductor (e.g., bulk silicon), a semiconductor-on-insulator (SOI) substrate, or the like, which may be doped (e.g., with a P-type or an N-type dopant) or undoped. The substrate **110** may be a wafer, such as a silicon wafer. Generally, an SOI substrate is a layer of a semiconductor material formed on an insulator layer. The insulator layer may be, for example, a buried oxide (BOX) layer, a silicon oxide layer, or the like. The insulator layer is provided on a substrate, typically a silicon or glass substrate. Other substrates, such as a multi-layered or gradient substrate may also be used. In some embodiments, the semiconductor material of the substrate **110** includes silicon, germanium, a compound semiconductor including silicon carbide, gallium arsenic, gallium phosphide, indium phosphide, indium arsenide, and/or indium antimonide, an alloy semiconductor including SiGe, GaAsP, AlInAs, AlGaAs, GaInAs, GaInP, and/or GaInAsP; or combinations thereof.

(13) In some embodiments, predetermined numbers of epitaxial layers may be formed on the semiconductor material of the substrate **110** through an epitaxial technique. A patterning process including an etching step is performed to remove a portion of the epitaxial layers and a portion of the semiconductor material of the substrate **110** to form trenches which define the semiconductor strip **112** and the epitaxial stack **120** over the semiconductor strip **112**. The etching may be any acceptable etch process, such as a reactive ion etch (RIE), neutral beam etch (NBE), the like, or a combination thereof. The etch may be anisotropic. After patterning the semiconductor strip **112**, the isolation structure **114** may be formed by filling dielectric material in the trenches which define the semiconductor strip **112**. The isolation structure **114** may be an oxide, such as silicon oxide, a nitride, the like, or a combination thereof, and may be formed by a high density plasma chemical vapor deposition (HDP-CVD), a flowable CVD (FCVD) (e.g., a CVD-based material deposition in a remote plasma system and post curing to make it convert to another material, such as an oxide), the like, or a combination thereof. Other insulation materials formed by any acceptable process may be used.

(14) The epitaxial stack **120** may include first epitaxial layers **122** and second epitaxial layers **124** alternately formed on the semiconductor strip **112** through an epitaxial growth process, which may be performed in a growth chamber. The first epitaxial layers **122** are formed of a first semiconductor material and the second epitaxial layer **124** are formed of a second semiconductor material that is different from the first semiconductor material. For example, the first semiconductor material is silicon germanium ($\text{Si}_{\text{sub.x}}\text{Ge}_{\text{sub.1-x}}$, where x can be in the range of 0 to 1), and the second semiconductor material is silicon. In some embodiments, the second epitaxial

layer **124** may be of the same material as the semiconductor strip **112**, but the disclosure is not limited thereto. The numbers of the first epitaxial layers **122** and the numbers of the second epitaxial layers **124** may be the same. In some embodiments, the numbers of the first epitaxial layers **122** or the second epitaxial layers **124** may be 1 to 5 layers. In addition, the top most layer of the epitaxial stack **120** may be the second epitaxial layer **124**.

(15) The epitaxial cap **130** may be formed on the epitaxial stack **120** over the semiconductor strip **112** through an epitaxial technique. The epitaxial cap **130** may include a first cap layer **132** and a second cap layer **134** sequentially formed on the epitaxial stack **120**. In some embodiments, the first cap layer **132** is selectively grown on exposed surfaces of the epitaxial stack **120** over the semiconductor strip **112** and the second cap layer **134** is selectively grown on exposed surfaces of the first cap layer **132**. The first cap layer **132** may be the same material as the first epitaxial layer **122**, for example silicon germanium ($\text{Si.sub.xGe.sub.1-x}$, where x can be in the range of 0 to 1) and the second cap layer **134** may be the same material as the second epitaxial layer **124**, for example silicon. A portion of the first cap layer **132** and a portion of the second cap layer **134** may be formed by laterally growing from the sidewall of the epitaxial stack **120** and may have a crystal orientation different from another portion of the first cap layer **132** and another portion of the second cap layer **134** that are vertically grown from the top of the epitaxial stack **120**.

(16) In some embodiments, the first cap layer **132** may be thicker than the second cap layer **134**. In some embodiments, the thickness of the second cap layer **134** may be from 1 nanometer to 5 nanometers. As shown in FIG. 1, the epitaxial cap **120** is conformal to the exposed surfaces of the epitaxial stack **120** and the semiconductor strip **112**. In addition, the first cap layer **132** and the second cap layer **134** may both disposed directly on the isolation structure **114**. In other words, the first cap layer **132** and the second cap layer **134** may both be in contact with the top surface of the isolation structure **114**.

(17) In FIG. 2, dielectric fins **140** are formed at a lateral side of the epitaxial cap **130** in the Y-axis. The dielectric fins **140** are formed by a low-K dielectric material on the isolation structure **114** of the substrate **110**. The low-K dielectric material may have a dielectric constant K smaller than about 7, such as smaller than about 3.9, and may include SiO.sub.2 , SiN , SiCN , or SiOCN . In some embodiments, the dielectric fins **140** may be in contact with the isolation structure **114** similar to the epitaxial cap **130**. In some embodiments, a planarization process, such as CMP, may be performed so that the top surface of the epitaxial cap **130** is revealed and the tops of the dielectric fin **140** and the epitaxial cap **130** are coplanar as shown in FIG. 2. In other words, the top of the epitaxial cap **130** may not be covered by the dielectric fins **140**.

(18) Thereafter, a removal process is performed to remove a portion of the epitaxial cap **130** from the exposed surface of the epitaxial cap **130** so that the epitaxial cap **130** is patterned to epitaxial sidewalls **150** as shown in FIG. 3. The removal process may include etching process that is able to etch back the epitaxial cap **130** until the top most layer, the second epitaxial layer **124**, of the epitaxial stack **120** is revealed. In FIG. 3, the height H_{150} of the epitaxial sidewalls **150** may be lower than the height H_{140} of the dielectric fin **140**. The top of the epitaxial stack **120** may be co-leveled to the top of the epitaxial sidewalls **150**. Specifically, the epitaxial sidewall **150** may include a first sidewall **152** that is formed from the first cap layer **132** and a second sidewall **154** that is formed from the second cap layer **134**. The top of the top most second epitaxial layer **124** of the epitaxial stack **120**, the top of the first sidewall **152** and the top of the second sidewall **154** are substantially co-leveled and lower than the top of the dielectric fin **140** to form a recess structure, but the disclosure is not limited thereto.

(19) In FIG. 4, a dummy gate structure **160** is formed on the structure of FIG. 3. The dummy gate structure **160** may include a gate dielectric **162** and a dummy gate electrode **164**. The gate dielectric **162** may be conformally form on tops of the dielectric fin **140**, the epitaxial sidewall **150** and the epitaxial stack **120**. The gate dielectric **162** may be, for example, silicon oxide, silicon nitride, multilayers thereof, or the like, and may be deposited or thermally grown. The dummy gate

electrode **164** may be formed on the gate dielectric **162** using the material, for example, polysilicon, or the like. The dummy gate structure **160** may be patterned to have a strip shape that extends along Y-axis and across the epitaxial stack **120** and the epitaxial sidewall **150** between the dielectric fins **140**. In other words, the extension direction of the dummy gate structure **160** may be substantially parallel to Y-axis. The dummy gate structure **160** may have a lengthwise direction substantially perpendicular to the lengthwise direction of the semiconductor strips **112** or the lengthwise direction of the dielectric fins **140**.

(20) After forming the dummy gate structure **160**, the step of forming a spacer structure, the step of forming the channel structure, the step of forming source/drain structures and the step of forming a replacement-gate (RPG) may be selectively performed. The spacer structure is disposed at opposite sides of the dummy gate structure **160** in the X-axis, and the source/drain structures are disposed at opposite sides of the epitaxial stack **120** and opposite sides of the epitaxial sidewall **150** in the X-axis, which are not presented in the cross section taken along the Y-axis. Therefore, the descriptions for the step of forming a spacer structure, the step of forming a source/drain structure may refer to other embodiments in this disclosure while be omitted in this embodiment. In addition, before the step of forming source/drain structures, the semiconductor strip **112**, the epitaxial stack **120** and the epitaxial sidewalls **150** are patterned so that the semiconductor strip **112**, the second epitaxial layers **124** of the epitaxial stack **120** and the second sidewalls **154** of the epitaxial sidewall **150** may be patterned to have a width in the X-axis similar to the width of the dummy gate structure **160**. The structures of the cross sections taken along X-axis with respect to the above steps may refer to other embodiments in the disclosure and be omitted in this embodiment showing the cross sections taken along Y-axis.

(21) In some embodiments, at the step of forming the channel structure, the semiconductor strip **112**, the epitaxial stack **120** and the epitaxial sidewalls **150** are patterned. In addition, the dummy gate electrode **164** may be removed, and then the first sidewalls **152** of the epitaxial sidewalls **150** and the first epitaxial layers **122** of the epitaxial stack **120** may be removed by a selective removal process. Therefore, the structure shown in FIG. 5 is obtained. Specifically, the semiconductor strip **112** of the substrate **110** is patterned to form a bottom channel structure **172**, the second epitaxial layers **124** are patterned to form embedded channel structures **174**, and the second epitaxial sidewalls **154** are patterned to form sidewall channel structures **176**. The removal of the first sidewalls **152** of the epitaxial sidewalls **150** and the first epitaxial layers **122** of the epitaxial stack **120** forms the gap VD. The bottom channel structure **172**, the embedded channel structures **174** and the sidewall channel structures **176** are exposed between the dielectric fins **140**, between the source/drawn structures (not shown and may refer to the embodiment showing the cross section in X-axis), and between spacer structures (not shown and may refer to the embodiment showing the cross section in X-axis).

(22) The step of forming PRG as shown in FIG. 6 includes forming a gate dielectric layer **180** and a gate electrode structure **190**. In some embodiments, the gate dielectric layer **180** and the gate electrode structure **190** may be known as a replacement gate structure. The gate dielectric layer **180** may include a dielectric portion **182** surrounding exposed surfaces of the embedded channel structures **174**, and a dielectric portion **184** continuously cover exposed surfaces of the dielectric fins **140**, the sidewall channel structures **176**, the isolation region **114**, and the bottom channel structure **172** as shown in FIG. 5. In accordance with some embodiments, the gate dielectric layer **180** includes silicon oxide, silicon nitride, or multilayers thereof. In some embodiments, the gate dielectric layer **180** is a high-k dielectric material, and in these embodiments, the gate dielectric layer **180** may have a k value greater than about 7.0, and may include a metal oxide or a silicate of Hf, Al, Zr, La, Mg, Ba, Ti, Pb, and combinations thereof. The formation methods of the gate dielectric layer **170** may include Molecular-Beam Deposition (MBD), ALD, PECVD, and the like.

(23) The gate electrode structure **190** fills the space between the dielectric fins **140**. Specifically, the gate electrode structure **190** is disposed between the lowest embedded channel structure **174L**

and the bottom channel structure **172**, between the lowest embedded channel structure **174L** and the adjacent embedded channel structure **174**, and between the embedded channel structures **174** and the sidewall channel structures **176**. In addition, the gate electrode structure **190** covers a portion of the dielectric layer **174** over the dielectric fin **140**. The gate electrode structure **190** may be a metal-containing material such as TiN, TiO, TaN, TaC, Co, Ru, Al, W, combinations thereof, or multi-layers thereof. The gate electrode structure **190** may have a multi-layered structure, and may comprise any number of liner layers, any number of work function layers, and a fill material that are stacked sequentially.

(24) In FIG. **6**, a semiconductor device **100** includes dielectric fins **140**, a bottom channel structure **172**, embedded channel structures **174**, sidewall channel structures **176**, a gate dielectric layer **180**, and a gate electrode structure **190**. The bottom channel structure **172** may be a protruded structure of the substrate **110** and the substrate **110** may further have an isolation structure **114** disposed beside the bottom channel structure **172** and substantially located at opposites of the bottom channel structure **172** in the direction of the Y-axis. The embedded channel structures **174** may be located above the bottom channel structure **172** and separated from the bottom channel structure **172** in the direction of Z-axis. The sidewall channel structures **176** are disposed on the sidewall of the dielectric fins **140**. The gate dielectric layer **180** covers and is in contact with the bottom channel structure **172**, the embedded channel structures **174**, and the sidewall channel structures **176**. The gate electrode structure **190** is disposed on the gate dielectric layer **180** and fill the space between two structures of the bottom channel structure **172**, the embedded channel structures **174**, and the sidewall channel structures **176**. In some embodiments, the gate dielectric layer **180** is disposed between the gate electrode structure **190** and the bottom channel structures **172**, between the gate electrode structure **190** and the embedded channel structures **174**, and between the gate electrode structure **190** and the sidewall channel structures **176**.

(25) As described in above, the substrate **110** has a semiconductor material and the bottom channel structure **172** is a protruded structure formed by a portion of the semiconductor material of the substrate **110**. In other words, the bottom channel structure **172** is formed in the substrate **110** and the isolation structure **114** laterally surrounds the bottom channel structure **172**. The isolation structure **114** may be an oxide that is deposited in the recess for defining the protruded structure of the bottom channel structure **172** (e.g. the semiconductor strip **112** shown in FIG. **1**). The bottom channel structure **172** may be higher than the isolation structure **114** and protruded from the isolation structure **114** in the direction of Z-axis.

(26) The embedded channel structures **174** may be disposed over the bottom channel structure **172** that is formed in the substrate **110** and the gate electrode structure **190** is disposed all around the embedded channel structures **174** so that the embedded channel structures **174** may be considered “embedded”. The embedded channel structures **174** may be aligned with the bottom channel structure **172** in the Z-axis. In some embodiments, the width **W174** of the embedded channel structures **174** may be substantially identical to the width **W172** of the bottom channel structure **172** in Y-axis. In some embodiments, the side surfaces of the embedded channel structures **174** and the side surface of the bottom channel structure **172** may extend on the same X-Z plane.

(27) The width **W174** of the embedded channel structures **174** may be greater than the height **H174** of the embedded channel structures **174**. The embedded channel structures **174** may be sheet-like structures extending in X-axis. In some embodiments, the width **W174** of the embedded channel structures **174** may be similar to the height **H174** of the embedded channel structures **174** and have wire-like structures extending in X-axis. In some embodiments, the embedded channel structures **174** may have a rectangular shape in the cross section while in alternative embodiments, the embedded channel structures **174** may have a circular or oval shape in the cross section, but the disclosure is not limited thereto.

(28) Two embedded channel structures **174** are presented in FIG. **6** for descriptive purpose. In some embodiments, the quantity of the embedded channel structures **174** is determined based on the

product requirement. For example, 1 to 5 embedded channel structures **174** may be formed in the semiconductor device **100**. The lower embedded channel structure **174L** is separated from the bottom channel structure **172** of the substrate **110** by a distance S_1 in the Z-axis, and the adjacent embedded channel structures **174** are separated from each other by a distance S_2 in the Z-axis. In some embodiments, the distance S_1 and the distance S_2 may be the same and may be determined according to the thickness of the above mentioned first epitaxial layers **122** that was removed during fabricating the semiconductor device **100**. The embedded channel structures **174** are made of semiconductor material such as silicon, but not limited thereto.

(29) The sidewall channel structures **176** may be wall-like structures disposed on the isolation structure **114** of the substrate **110** and located at a lateral side of the embedded channel structures **174** in the Y-axis. The sidewall channel structure **176** is disposed between the gate dielectric layer **180** and the dielectric fin **140**. The sidewall channel structures **176** are directly disposed on and in contact with the sidewall of the dielectric fins **140**. The sidewall channel structure **176** may have an elongate shape extending in Z-axis in the cross section of FIG. 6. In some embodiments, the width W_{176} of the sidewall channel structure **176** may be 1 nanometer to 5 nanometers and the height H_{176} of the sidewall channel structure **176** may be 15 nanometers to 60 nanometers. The sidewall channel structure **176** may also extend in X-axis to form a wall-like structure along the sidewall of the dielectric fins **140**.

(30) The sidewall channel structures **176** are disposed on the isolation layer **114**, located at opposite sides of the embedded channel structures **174** in Y-axis and separated from the embedded channel structures **174** by a distance S_3 . Therefore, the bottom channel structure **172**, the embedded channel structures **174** and the sidewall channel structures **176** are separate channel structures. In some embodiments, the distance S_3 may be determined based on the thickness of the above mentioned first cap layer **132** that was removed during fabricating the semiconductor device **100**. The sidewall channel structures **176** are patterned from sidewall portions of the second cap layer **134** in FIG. 1 and the sidewall portion of the second cap layer **134** in FIG. 1 may be laterally grown from the sidewall portion of the first cap layer **132** in FIG. 1. In some embodiments, the crystal orientation of the sidewall channel structures **176** may be different from the crystal orientation of the embedded channel structure **174** since the embedded channel structure **174** is formed from the second epitaxial layer **124** in FIG. 1 that is vertically grown from the first epitaxial layer **122** in FIG. 1.

(31) The dielectric fins **140** are disposed on the isolation structure **114** of the substrate **110** and separated by a gap G_{140} . The embedded channel structures **174** and the sidewall channel structures **176** are disposed within the gap G_{140} . The sidewall channel structures **176** extends along and physically contact the sidewall of the dielectric fins **140**. The height H_{140} of the dielectric fin **140** may be higher than the height H_{176} of the sidewall channel structure **176**. In some embodiments, the height H_{140} of the dielectric fins **140** may be 20 nanometers to 65 nanometers and the width W_{140} of the dielectric fins **140** may be 3 nanometers to 50 nanometers, but the disclosure is not limited thereto. In addition, the top of the top most embedded channel structure **174T** may be leveled to the tops of the sidewall channel structures **176** while the tops of the dielectric fins **140** are leveled higher than the top most embedded channel structure **174T** and the sidewall channel structures **176**. The dielectric fins **140** are made of a low-K dielectric material such as SiO_2 , SiN, SiCN, or SiOCN.

(32) The gate dielectric layer **180** includes a dielectric portion **182** and a dielectric portion **184**. The dielectric portion **182** and the dielectric layer **184** are made of a high-k dielectric material and may include a metal oxide or a silicate of Hf, Al, Zr, La, Mg, Ba, Ti, Pb, and combinations thereof. The dielectric portion **182** is disposed on the embedded channel structures **174** and substantially wraps the embedded channel structures **174** to form a ring-like pattern in the cross section of FIG. 6. The dielectric portion **184** is disposed on the bottom channel structure **172** and the sidewall channel structures **176**. Specifically, the dielectric portion **184** may be conformally disposed on and cover

the bottom channel structure **172**, the sidewall channel structures **176**, a portion of the isolation structure **114** between the bottom channel structure **172** and the sidewall channel structures **176**, and the dielectric fin **140**. Accordingly, the bottom channel structure **172**, the embedded channel structure **174** and the sidewall channel structure **176** are covered by the gate dielectric layer **180**. (33) The gate electrode structure **190** at least fills the space between the dielectric portion **182** and the dielectric portion **184**. Specifically, the gate electrode structure **190** encircles the embedded channel structures **174** and is located between the embedded channel structures **174** and the sidewall channel structures **176**. The gate electrode structure **190** includes metal-containing material such as TiN, TiO, TaN, TaC, Co, Ru, Al, W, combinations thereof, or multi-layers thereof. The gate electrode structure **190** may have a multi-layered structure, and may comprise any number of liner layers, any number of work function layers, and a fill material that are stacked sequentially. In addition, the gate electrode structure **190** is isolated from the bottom channel structure **172**, the embedded channel structures **174** and the sidewall channel structures **176** by the gate dielectric layer **180**. In addition, the gate electrode structure **190** may have a sufficient thickness in the X-axis so as to contact to other components for signal transmission. In other words, the top of the gate electrode structure **190** may be higher than the gate dielectric layer **180** in Z-axis.

(34) The electric signal applied to the gate electrode structure **190** enables or disables the bottom channel structure **172**, the embedded channel structures **174** and the sidewall channel structures **176** so as to achieve the switch of the semiconductor device **100**. In other words, the voltage value on the gate electrode structure **190** may change the carrier mobility of the bottom channel structure **172**, the embedded channel structures **174** and the sidewall channel structures **176** to present turn-on or turn-off status. The bottom channel structure **172**, the embedded channel structures **174** and the sidewall channel structures **176** all serve as the switchable channels of the semiconductor device **100**. The channel size of the semiconductor device **100** may be increased by disposing the bottom channel structure **172**, the embedded channel structures **174** and the sidewall channel structures **176**. The semiconductor device **100** having increased channel size allows to provide more driving current to achieve higher performance. In addition, the sidewall channel structures **176** are thin wall-like structures that occupies a limited volume so that the semiconductor device **100** is still compact in physical size and is able to be arranged in high density in the product. Therefore, the semiconductor device **100** has a larger channel while remains a small volume.

(35) In some embodiments, the embedded channel structures **174** and the sidewall channel structures **176** are formed from the second epitaxial layer **124** and the second cap layer **134** in FIG. 1. The second epitaxial layer **124** for forming the embedded channel structures **174** and the second cap layer **134** for forming the sidewall channel structures **176** may be grown in different directions, for example, the vertical direction and the lateral direction. In some embodiments, the second epitaxial layer **124** for forming the embedded channel structures **174** grown in the vertical direction may have a crystal orientation of **(100)** and the second cap layer **134** for forming the sidewall channel structures **176** grown in the lateral direction may have a crystal orientation of **(110)**. Therefore, the embedded channel structures **174** and the sidewall channel structures **176** may provide different properties for different carrier. In some embodiments, the sidewall channel structures **176** having the crystal orientation of **(110)** may further enhance the carrier mobility for a P-type carrier. Therefore, in the case the semiconductor device **100** is a P-type device, the carrier mobility may be further enhanced.

(36) FIG. 7 to FIG. 12 schematically illustrate a method of fabricating a semiconductor device from a plane view in accordance with some embodiments of the disclosure. Referring to the X-axis, the Y-axis and the Z-axis, FIG. 7 to FIG. 12 depict the plane views of the structures cut along line I-I' of FIG. 6 under respective steps. The same reference numbers in the embodiment of FIG. 1 to FIG. 6 and the embodiment of FIG. 7 to FIG. 12 may present the same elements or their alternatives and the descriptions for these elements in different embodiments may be applicable to each other.

(37) FIG. 7 shows the second epitaxial layer **124** is disposed between two portions of the first cap layer **132** and the two portions of the first cap layer **132** are disposed between two portions of the second cap layer **134**. In addition, the first cap layer **132** and the second cap layer **134** construe the epitaxial cap **130** as shown in FIG. 1 and the second epitaxial layer **124** is one layer of the epitaxial stack **120** as shown in FIG. 1. Specifically, FIG. 7 may present the plane view of the structure under the step of FIG. 1. The epitaxial stack **120** may be corresponding to the semiconductor strip **112** located at a level lower than the line I-I' of FIG. 6 and thus the semiconductor strip **112** is presented by dash line in FIG. 7. In some embodiments, the portions of the first cap layer **132** in FIG. 7 may be laterally grown on the side wall of the epitaxial stack **120** and the portions of the second cap layer **134** may be laterally grown on the sidewall of the first cap layer **132**. The material of the first cap layer **132** may be silicon germanium and the material of the second cap layer **134** may be silicon. The material of the second epitaxial layer **124** may be the same as the material of the second cap layer **134**. The second epitaxial layer **124**, the first cap layer **130**, and the second cap layer **134** may extend in X-axis.

(38) In FIG. 8, the dielectric fins **140** are formed beside the epitaxial cap **130** and at two opposite sides of the epitaxial stack **124** in the Y-axis. The dielectric fins **140** are made of a low-K dielectric material and may include SiO.sub.2, SiN, SiCN, or SiOCN. The first cap layer **132**, the second cap layer **134** and the dielectric fins **140** are disposed beside the second epitaxial layer **124** and sequentially arranged away from the second epitaxial layer **124** in a lateral direction in Y-axis. In some embodiments, FIG. 8 may present the plane view of the structure under the step of FIG. 2 and the step of FIG. 3.

(39) In FIG. 9, the step of forming the channel structure is performed and the step of FIG. 9 may be included in the step of FIG. 4. The embedded channel structures **174** and the sidewall channel structures **176** are formed by patterning the epitaxial stack **120** and the epitaxial cap **130** in FIG. 1. Specifically, as shown in FIG. 9, the second epitaxial layer **124** is patterned to the embedded channel structure **174**, the first cap layer **132** is patterned to form a shrunk first cap layer **132'**, and the second cap layer **134** is patterned to the sidewall channel structure **176**. In addition, a portion of the semiconductor strip **112** is patterned to form the bottom channel structure **172**. In FIG. 9, a selective removal process may be performed so that the first cap layer **132** is patterned to the shrunk first cap layer **132'** that is shrunk relative to the embedded channel structure **174** and the sidewall channel structure **176**.

(40) As shown in FIG. 9, measured along the X-axis, the length L₁₄₀ of the dielectric fins **140** is greater than the length L₁₇₆ of the sidewall channel structure **176**, the length L₁₇₆ of the sidewall channel structure **176** is greater than the length L_{132'} of the shrunk first cap layer **132'**, and the length L₁₇₄ of the embedded channel structure **174** is greater than the length L_{132'} of the shrunk first cap layer **132'**. In some embodiments, the length L₁₇₆ of the sidewall channel structure **176** may be substantially the same as the length L₁₇₄ of the embedded channel structure **174** so as the bottom channel structure **172**. In addition, the width W₁₇₄ of the embedded channel structure **174** is smaller than the length L₁₇₄ of the embedded channel structure **174** and the width W₁₇₆ of the sidewall channel structure **176** is smaller than the length L₁₇₆ of the sidewall channel structure **176**, but the disclosure is not limited thereto.

(41) In FIG. 10, the step of forming a spacer structure and the step of forming source/drain structures are performed. Specifically, inner spacers **210** may be formed to fill the space left by the removal (e.g., recess formed by the shrunk first cap layer **132'**) of the first cap layer **132** discussed above with reference to FIG. 9. The inner spacers **210** may be a low-K dielectric material, such as SiO.sub.2, SiN, SiCN, or SiOCN, and may be formed by a suitable deposition method, such as ALD. The inner spacers **210** are formed to fill the recess structure of the shrunk first cap layer **132'** without covering the embedded channel structure **174** and the sidewall channel structure **176**. Therefore, the side surfaces of the inner spacer **210**, the embedded channel structure **174** and the sidewall channel structure **176** may be aligned as shown in FIG. 10.

(42) In FIG. 10, source/drain structures 220 are formed at lateral sides of the embedded channel structure 174 and the sidewall channel structures 176 in X-axis. The source/drain structures 220 may be formed by epitaxially growing a material over the semiconductor strip 112, the bottom channel structure 172, the embedded channel structures 174 and the sidewall channel structures 176, using suitable methods such as metal-organic CVD (MOCVD), molecular beam epitaxy (MBE), liquid phase epitaxy (LPE), vapor phase epitaxy (VPE), selective epitaxial growth (SEG), the like, or a combination thereof. The semiconductor strip 112 is located at a level lower than the line I-I' of FIG. 6 and thus is presented by dash line in FIG. 10. The source/drain structures 220 may be formed between the dielectric fins 140 and located at opposite sides of the channel structure construed by the embedded channel structure 174 and the sidewall channels 176 in X-axis. In addition, the inner spacer 210 is disposed between the shrunk first cap layer 132' and the source/drain structures 220. The material(s) of the source/drain structures 220 may be tuned in accordance with the type of devices to be formed. In some embodiments, the resulting semiconductor device is an n-type FinFET, and the source/drain structures 220 include silicon carbide (SiC), silicon phosphorous (SiP), phosphorous-doped silicon carbon (SiCP), or the like. In some embodiments, the resulting semiconductor device is a p-type FinFET, and the source/drain structures 220 include SiGe, and a p-type impurity such as boron or indium.

(43) Thereafter, the shrunk first cap layer 132' are removed to obtain the structure shown in FIG. 11. In some embodiments, referring to FIG. 4 and FIG. 5, the dummy gate structure 160 on the first cap layer 132 may be removed in advance to reveal the top surface of the shrunk first cap layer 132' and then a selective removal process may be performed to remove the shrunk first cap layer 132' without removing the embedded channel structure 174. Accordingly, as shown in FIG. 5 and FIG. 11, the bottom channel structure 172, the embedded channel structure 174 and the sidewall channel structure 176 are separated from each other and exposed by the gap VD. The bottom channel structure 172, the embedded channel structure 174 and the sidewall channel structure 176 may be connected to each other through the source/drain structures 220. The source/drain structures 220 are not revealed by the gap VD of removing the shrunk first cap layer 132' since the shrunk first cap layer 132' is isolated from the source/drain structures 220 by the inner spacer 210 as shown in FIG. 10.

(44) In FIG. 12, the step of forming replacement gate is performed. Specifically, the gate dielectric structure 180 and the gate electrode structure 190 are formed in sequence in the gap VD between bottom channel structure 172, the embedded channel structure 174 and the sidewall channel structure 176. The gate dielectric structure 180 may include the dielectric portion 182 covering and contacting the sidewall channel structure 174, the dielectric portion 184 covering and contacting the embedded channel structure 174 and the dielectric portion 186 between the dielectric portion 182 and the dielectric portion 184. The dielectric portion 186 may cover the inner spacer 210. Accordingly, the gate dielectric structure 180 is a continuous layer covering the bottom channel structure 172, the embedded channel structure 174, the sidewall channel structure 176 and the inner spacer 210 and forms a ring-like pattern in the plane view of FIG. 12.

(45) The gate electrode structure 190 is formed on the gate dielectric structure 180 so that the gate electrode structure 190 is isolated from the bottom channel structure 172, the embedded channel structure 174 and the sidewall channel structure 176 by the gate dielectric structure 180. The material and the manufacturing method of the gate electrode structure 190 may refer to the previous embodiment and is not reiterated here.

(46) In FIG. 12, a semiconductor device 200 may have the plane view structure including the embedded channel structure 174, the sidewall channel structure 176, the dielectric fins 140, the inner spacer 210, the source/drain structures 220, the gate dielectric structure 180 and the gate electrode structure 190. The source/drain structures 220 are disposed at opposite sides of the embedded channel structure 174 and the third structures 176 in the X-axis. The gate electrode structure 190 is separated from the embedded channel structure 174 and the sidewall channel

structures **176** by the gate dielectric layer **180** and separated from the source/drain structures **220** by the gate dielectric layer **180** and the inner spacer **210**. For example, the gate electrode structure **190** is laterally surrounded by the inner spacer **210**, the embedded channel structure **174** and the sidewall channel structure **176**. The inner spacers **210** and the gate electrode structure **190** are disposed between the embedded channel structure **174** and the sidewall channel structure **176** in the Y-axis. In addition, the side surface **S210** of the inner spacers **210**, the side surface **S174** of the embedded channel structure **174** and the side surface **S176** of the sidewall channel structure **176** are co-planar and are in contact with the source/drain structure **220**. In some embodiments, the cross section shown in FIG. **6** may be corresponding to the line V-V' in FIG. **12**.

(47) The semiconductor device **200** have various types of channel structures which facilitates to improve the design variety of the device. The length **L174** of the embedded channel structure **174** may be equivalent to the length **L176** of the sidewall channel structure **176**, and the length **L174** and the length **L176** may be determined based on the required characteristic of the semiconductor device **200**. The disposition of the sidewall channel structure **176** increases the channel size of the semiconductor device **200** without increase the dimension of the semiconductor device **200** in X-axis. In addition, the width **W176** of the sidewall channel structure **176** may be small, e.g. 1 nanometer to 5 nanometers so that the disposition of the sidewall channel structures **176** would increase a small amount of the device size in Y-axis, which facilitates the application in a high device density product. In some embodiments, the embedded channel structure **174** and the sidewall channel structures **176** may be grown in different epitaxial direction so as to have different crystal orientations, which may further enhance the performance of specific type device. For example, the channel structure grown in the crystal orientation of **(110)** enhances the performance of P-type device.

(48) FIG. **13** to FIG. **20** schematically illustrate a method of fabricating a semiconductor device from a cross section in accordance with some embodiments of the disclosure. Referring to the X-axis, the Y-axis and the Z-axis, FIG. **13** to FIG. **20** depict the cross sections of the structures cut along line II-II' of FIG. **6** under respective steps. The same reference numbers in the embodiment of FIG. **13** to FIG. **20** and the previous embodiments may present the same elements or their alternatives and the descriptions for these elements in different embodiments may be applicable to each other.

(49) In FIG. **13**, the epitaxial stack **120** and the epitaxial cap **130** are formed on the semiconductor strip **112**. The epitaxial stack **120** includes the first epitaxial layers **122** and the second epitaxial layers **124** stacked alternatively on the semiconductor strip **112**. The epitaxial cap **130** includes the first cap layer **132** and the second cap layer **134** sequentially formed on the epitaxial stack **120**. Similar to the previous embodiments, the first epitaxial layers **122** and the first cap layer **132** may be formed of a material different from the second epitaxial layer **124** and the second cap layer **134**. For example, the material of the first epitaxial layers **122** and the first cap layer **132** may be silicon germanium and the material of the second epitaxial layers **124** and the second cap layer **134** may be silicon.

(50) In FIG. **14**, the epitaxial cap **130** above the epitaxial stack **120** is removed and the dummy gate structure **160** is formed on the epitaxial stack **120**, which may be corresponding to the step of FIG. **3** and the step of FIG. **4**. The dummy gate structure **160** may include a gate dielectric **162** and a dummy gate electrode **164**. The gate dielectric **162** may be formed between the dummy gate electrode **164** and the epitaxial stack **120**. In some embodiments, the dummy gate structure **160** is formed on the top most second epitaxial layer **124** of the epitaxial stack **120**.

(51) In FIG. **15**, a spacer **310** is formed beside the dummy gate structure **160**. In some embodiments, a spacer material layer (not shown) may be silicon nitride, silicon carbonitride, a combination thereof, or the like. In some embodiments, the spacer material layer includes multiple sublayers. For example, a first sublayer (sometimes referred to as a gate seal spacer layer) may be formed by thermal oxidation or a deposition, and a second sublayer (sometimes referred to as a

main gate spacer layer) may be conformally deposited on the first sublayer. The spacer material may be patterned by an anisotropic etching to form the spacer **310**.

(52) After forming the spacer **310**, an anisotropic etching process is performed to form a shorten epitaxial stack **120'** including the embedded channel structures **174** and the shorten first epitaxial layer **122'**. Specifically, the embedded channel structures **174** from the second epitaxial layers **124** and form the shorten first epitaxial layers **122'** from the first epitaxial layer **122**. In addition, a portion of the semiconductor strip **112** under the shorten epitaxial stack **120'** may be defined as the bottom channel structure **172**. In some embodiments, the anisotropic etching process is a dry etch process (e.g., a plasma etch process). In some embodiments, the length **L174** of the embedded channel structures **174** may be corresponding to the dimension **L310** of the formed structure of the dummy gate structure **610** and the spacer **310**. The sidewalls of the embedded channel structures **172** and the first epitaxial layers **122'** may be aligned with the sidewall of the spacer **310**.

(53) In FIG. **16**, a lateral etching process is performed to recess exposed portions of the first semiconductor material of the shorten first epitaxial layers **122'** using an etchant that is selective to the first semiconductor material while the second semiconductor material of the embedded channel structures **174** is not removed. Therefore, the shrunk first epitaxial layers **122''** as well as the shrunk first cap layer **132'** in FIG. **9** is formed and a recess structure is formed by the first epitaxial layers **122''** and the embedded channel structures **174**.

(54) In FIG. **17**, the inner spacer **210** and the source/drain structures **220** are formed, similar to the step of FIG. **10**. In some embodiments, the steps of FIG. **14** to FIG. **17** may be corresponding to the step of FIG. **4** and may be references for the step of forming the spacer, the step forming the channel structure and the step of forming source/drain structures. The side surface **S210** of the inner spacers **210**, the side surface **S174** of the embedded channel structure **174** and the side surface **S310** of the spacer **310** may be co-planar. The source drain structures **220** are formed at two opposite lateral sides of the embedded channel structures **174** and in contact with the side surface **S174** and the side surface **S210**. In FIG. **18**, an interlayer dielectric (ILD) layer **320** may be formed on the source/drain structures **220** at the lateral side the spacers **310** in X-axis. The ILD layer **320** includes a dielectric material such as silicon oxide, phosphosilicate glass (PSG), borosilicate glass (BSG), boron-doped phosphosilicate Glass (BPSG), undoped silicate glass (USG), or the like, and may be deposited by any suitable method, such as CVD, PECVD, or FCVD. In addition, a removal step similar to the step of FIG. **5** and the step of FIG. **11** is performed so that the shrunk first epitaxial layers **122''** and the shrunk first cap layer **132'** (shown in FIG. **10**) are removed to provide the gap **VD** between the bottom channel structure **172**, the embedded channel structures **174** and the sidewall channel structures **176** to expose the inner surfaces **I310** of the spacer **310** and the inner surfaces **1210** of the inner spacers **210**. The gap **VD** between the embedded channel structures **174** shown in FIG. **18** may extend along the embedded channel structures **174** between the embedded channel structure **174** and the sidewall channel structure **176** in Y-axis and communicated with gap **VD** shown in FIG. **11**.

(55) Subsequently, the step of forming the gate dielectric layer **180** and the gate electrode structure **190** similar to the step of FIG. **12** is performed as shown in FIG. **19**. The gate dielectric layer **180** and the gate electrode structure **190** fill the gap **VD** between the spacer **310**, the inner spacers **210**, the bottom channel structure **172**, the embedded channel structures **174** and the sidewall channel structures **176**. The gate dielectric layer **180** separates the gate electrode structure **190** from the spacer **310**, the inner spacers **210**, the bottom channel structure **172**, the embedded channel structures **174** and the sidewall channel structures **176**. In FIG. **20**, contact metals **330** may be formed in the ILD layer **320** and in contact with the source/drain structures **220**. In some embodiments, a planarization process may be subsequently performed so that the tops of the contact metals **330**, the gate electrode structure **190**, the gate dielectric layer **190** and the spacer **310** are substantially at the same level.

(56) A semiconductor device **300** in FIG. **20** includes the bottom channel structure **172** formed in

the semiconductor strip **112** of the substrate **110**, the source/drain structures **220**, the embedded channel structures **174** disposed between the source/drain structures **220**, the gate dielectric layer **180**, the gate electrode structures **190**, the inner spacers **210** and the spacer **310** beside the gate electrode structure **190**, the ILD layer **320** and the contact metals **330**. The embedded channel structures **174** may have sheet-like structure extending along the X-axis and connecting to the source/drain structures **220** at the opposite sides in the X-axis. The inner spacers **210** and the gate electrode structure **190** are disposed between adjacent embedded channel structures **174** while the gate dielectric layer **180** is disposed between the gate electrode structure **190** and the embedded channel structure **174**. The gate dielectric layer **180** is further disposed between the gate electrode structure **190** and the bottom channel structure **172**, between the gate electrode structure **190** and the inner spacers **210**, and between the gate electrode structure **190** and the spacers **310**. The source/drain structures **220** are formed and disposed directly on the semiconductor strip **112** of the substrate **110**. The ILD layer **320** is disposed on the source/drain structures **220** and the metal contacts **320** extend through the ILD layer **320** to contact the source/drain structures **220**. A planarization process may be performed on the semiconductor device **300** so that the top surface **T190** of the gate electrode structure **190**, the top surface **T180** of the gate dielectric layer **180**, the top surface **T310** of the spacer **310**, the top surface **T320** of the ILD layer **320** and the top surface **T330** of the metal contact **330** are coplanar.

(57) FIG. **21** to FIG. **27** schematically illustrate a method of fabricating a semiconductor device from a cross section in accordance with some embodiments of the disclosure. Referring to the X-axis, the Y-axis and the Z-axis, FIG. **21** to FIG. **27** depict the cross sections of the structures cut along line III-III' of FIG. **6** under respective steps. The same reference numbers in the embodiment of FIG. **21** to FIG. **27** and the previous embodiments of FIG. **1** to FIG. **20** may present the same elements or their alternatives and the descriptions for these elements in different embodiments may be applicable to each other.

(58) In FIG. **21**, the second cap layer **134** is disposed on the isolation structure **114** of the substrate **110**, which is similar to the steps of FIG. **1**, FIG. **7** and FIG. **13** of the previous embodiments. The second cap layer **134** is formed on the isolation structure **114** using the epitaxial technique and is made of a semiconductor material such as silicon. In FIG. **22**, the dummy gate structure **160** and the spacer **310** are disposed on the second cap layer **134**. The dummy gate structure **160** may be formed by the method similar to that described in FIG. **4** and FIG. **14**. The dummy gate structure **160** may include the gate dielectric **162** and the dummy gate electrode **164**. The spacer **310** is formed beside the dummy gate structure **160** and disposed at opposite sides of the dummy gate structure **160** in the x-axis. The formation of the spacer **310** may relate to the description of FIG. **15**.

(59) In FIG. **23**, a patterning process is performed to form the sidewall channel structure **176** from the second cap layer **134** of FIG. **22**. In some embodiments, the side surface **S176** of the sidewall channel structure **176** may be aligned with the side surface **S310** of the spacer **310**. The steps of FIG. **22** and FIG. **23** may be similar to the step of FIG. **15**. In some embodiments, the length **L176** of the embedded channel structures **174** may be corresponding to the dimension **L310** of the formed structure of the dummy gate structure **610** and the spacer **310**, when measured along the X-axis. In FIG. **24**, the source/drain structures **220** are formed at the side surface **S176** of the sidewall channel structure **176** and the step of FIG. **24** may be similar to the step of FIG. **10** and the step of FIG. **17**. In some embodiments, the step of FIG. **16** and the step of forming the inner spacers **210** depicted in FIG. **17** may be performed between the step of FIG. **23** and the step of FIG. **24**.

(60) In FIG. **25**, the interlayer dielectric (ILD) layer **320** is formed at the lateral side of the spacer **310** in the X-axis and the dummy gate structure **160** is removed. The ILD layer **320** includes a dielectric material such as silicon oxide, phosphosilicate glass (PSG), borosilicate glass (BSG), boron-doped phosphosilicate Glass (BPSG), undoped silicate glass (USG), or the like. The step of FIG. **25** is similar to the step of FIG. **5**, the step of FIG. **11** and the step of FIG. **18**. The top surface

T176 of the sidewall channel structure **176** may be exposed due to the removal of the dummy gate structure **160** and the gap **VD** between the spacers **310** is provided. In FIG. **26**, the gate dielectric layer **180** and the gate electrode structure **190** are sequentially formed in the gap **VD** between the spacers **310**. The step of FIG. **26** is similar to the step of FIG. **6**, the step of FIG. **12** and the step of FIG. **19**. The materials, the forming method and the disposition of the gate dielectric layer **180** and the gate electrode structure **190** may refer to the previous embodiments. In FIG. **27**, the contact metals **330** may be formed in the ILD layer **320** and in contact with the sidewall channel structure **220**. Specifically, the contact metals **330** extend through the ILD layer **320** in the Z-axis so that the top surface **T330** of the contact metals **330** may be co-planar to the top surface **T320** of the ILD layer **320** and the bottom **B330** of the contact metals **330** may be in contact with the sidewall channel structure **220**. The step of FIG. **27** may be similar to the step of FIG. **20**.

(61) In FIG. **27**, a semiconductor device **400** includes the isolation structure **114**, the sidewall channel structure **176**, the source/drain structures **220**, the gate electrode structure **190**, the gate dielectric layer **180**, the spacer **310**, the ILD layer **320** and the metal contacts **330**. The sidewall channel structure **176** and the source/drain structures **220** are disposed on the isolation structure **114**. The source/drain structures **220** are located at and in contact with opposite side surfaces **S176** of the sidewall channel structure **176** in the X-axis. The gate electrode structure **190** is disposed over the sidewall channel structure **176** and the gate dielectric layer **180** is disposed between the gate electrode structure **190** and the sidewall channel structure **176**. The spacers **310** are disposed on the sidewall channel structure **176** and located at opposite sides of the gate electrode structure **190** in the X-axis. The outer surface **S310** of spacers **310** away from the gate electrode structure **190** may be aligned with the sidewall **S176** of the sidewall channel structure **176**. The metal contacts **330** are disposed on the source/drain structures **220** and laterally surrounded by the ILD layer **320**. The top surface **T320** of the ILD layer **320**, the top surface **T330** of the metal contacts **330**, the top surface **T310** of the spacers **310**, the top surface **T190** of the gate electrode structure **190** and the top surface **T180** of the gate dielectric layer **180** may be co-planar.

(62) FIG. **28** to FIG. **35** schematically illustrate a method of fabricating a semiconductor device from a cross section in accordance with some embodiments of the disclosure. Referring to the X-axis, the Y-axis and the Z-axis, FIG. **28** to FIG. **35** depict the cross sections of the structures cut along line IV-IV' of FIG. **6** under respective steps. The same reference numbers in the embodiment of FIG. **28** to FIG. **35** and the previous embodiments may present the same elements or their alternatives and the descriptions for these elements in different embodiments may be applicable to each other.

(63) In FIG. **28**, the first cap layer **132** is formed on the isolation structure **114** of the substrate **110**. The step of FIG. **28** may be similar to the step of FIG. **1**, the step of FIG. **7**, the step of FIG. **13** and the step of FIG. **21**. In FIG. **29**, the dummy gate structure **160** including the gate dielectric **162** and the dummy gate structure **164** is formed on the first cap layer **132** by a method similar to the step of FIG. **4**, the step of FIG. **14**, and the step of FIG. **22**. In some embodiments, the step of FIG. **2** and the step of FIG. **3** may be performed between the step of FIG. **28** and the step of FIG. **29**. In FIG. **30**, the spacer **310** is formed and a shorten first cap layer **132''** is formed from the first cap layer **132**. Specifically, the spacer **310** may be formed on the first cap layer **132** of FIG. **29** and a patterning process is performed on the first cap layer **132** to form the shorten first cap layer **132''**. In some embodiments, the side surface **S132''** of the shorten first cap layer **132''** may be aligned with the side surface **S310** of the spacers **310** away from the dummy gate structure **160**. The step of FIG. **30** may be similar to the step of FIG. **15** and the step of FIG. **23**. Thereafter, the shorten first cap layer **132''** is further patterned to form the shrunk first cap layer **132'** as shown in FIG. **31**. The step of FIG. **31** may be similar to the step of FIG. **9** and the step of FIG. **16**. In some embodiments, the width of the shrunk first cap layer **132'** may be the same as the dummy gate structure **160**, but the disclosure is not limited thereto. In FIG. **32**, the inner spacers **210** and the source/drain structures **220** are sequentially formed on the isolation structure **114** of the substrate **110**. The inner

spacers **210** may be formed at the lateral sides of the shrunk first cap layer **132'** and the source/drain structures **220** are formed at the outer side surfaces **S210** of the inner spacers **220** using the method similar to the step of FIG. **10** and the step of FIG. **17**. The inner spacers **210** is disposed between the shrunk first cap layer **132'** and the source/drain structures **220**. The inner spacers **210** may be connected to the spacers **310** to form a continuous spacer. In addition, the side surface **S210** of the inner spacers **210** may be aligned with the side surface **S310** of the spacer **310**. (64) Thereafter, the dummy gate structure **160** and the shrunk first cap layer **132'** are removed in sequence, which is similar to the step of FIG. **5**, the step of FIG. **11**, the step of FIG. **18**, and the step of FIG. **28**. The gap **VD** ash shown in FIG. **33** is provided due to the removal of the dummy gate structure **160** and the shrunk first cap layer **132'**. In FIG. **34**, the ILD layer **320** is also formed to laterally surround the spacers **310**. In addition, the gate dielectric layer **180** and the gate electrode structure **190** are formed to fill the gap **VD**, which are similar to the step of FIG. **6**, the step of FIG. **12**, the step of FIG. **19** and the step of FIG. **29**. In FIG. **35**, the metal contacts **330** are formed extending through the ILD layer **320** and in contact with the source/drain structures **220**. (65) A semiconductor device **500** as shown in FIG. **35** includes the isolation structure **114** of the substrate **110**, the source/drain structures **220**, the inner spacers **210**, the gate electrode structure **190**, the gate dielectric layer **180**, the spacers **310**, the ILD layer **320** and the metal contact **330**. The source/drain structures **220**, the inner spacers **210**, and the gate dielectric layer **180** are disposed on the isolation structure **114** of the substrate **110**. The gate electrode structure **190** is disposed on the gate dielectric layer **180**. The spacers **310** is disposed on the inner spacers **210**. The metal contacts **330** and the ILD layer **320** are disposed on the source/drain structures **220**. The spacers **310** and the inner spacers **210** may be aligned and disposed beside the gate electrode structure **190**. The gate dielectric layer **180** may be conformally disposed along the inner surfaces of the inner spacers **210** and the inner surfaces of the spacers **310**. In some embodiments, the dimension of the inner spacers **210** and the dimension of the spacers **310** in the X-axis may be different while the gate dielectric layer **180** may be still conformally disposed along the inner surfaces of the inner spacers **210** and the inner surfaces of the spacers **310**. The gate dielectric layer **180** is disposed to separate the gate electrode structure **190** from the inner spacers **210**, the spacers **310**, and the isolation structure **114** of the substrate **110**. In addition, the tops of ILD layer **320**, the metal contacts **330**, the spacers **310**, the gate dielectric layer **180** and the gate electrode structure **190** may be coplanar through a planarization process.

(66) In some embodiments of the disclosure, the semiconductor device may have various shapes of channel structures to increase the channel size. For example, the semiconductor device may include the bottom channel structure formed from the protruded structure of the substrate, the embedded channel structures having sheet-like or wire-like structures, and the sidewall channel structures having wall-like structures. The sidewall channel structures having the wall-like structures requires a limited space so that the disposition of the sidewall channel structure involves the effect of increasing the channel size without significantly enlarge the device size. In some embodiments, the wall-like sidewall channel structure and the sheet-like embedded channel structure may have different crystal orientation so that the electric property of the semiconductor device may be enhanced. In the case that the channel structure in a p-type device has the crystal orientation of **(110)**, the carrier mobility may be further enhanced.

(67) In accordance with some embodiments, a semiconductor device may include an embedded channel structure disposed on a substrate; a sidewall channel structure disposed on the substrate, and located at a lateral side of the embedded channel structure; and a gate electrode structure disposed on the substrate, encircling the embedded channel structure and located between the embedded channel structure and the sidewall channel structure. In accordance with some embodiments, the embedded channel structure and the sidewall channel structure have different crystal orientations. In accordance with some embodiments, a dielectric fin is further disposed on the substrate, wherein the sidewall channel structure is in contact with a sidewall of the dielectric

fin. In accordance with some embodiments, a gate dielectric layer is further disposed between the gate electrode structure and the sidewall channel structure, wherein the sidewall channel structure is disposed between the gate dielectric layer and the dielectric fin. In accordance with some embodiments, the embedded channel structure is separated from the substrate by a distance. In accordance with some embodiments, a bottom channel structure is further formed in the substrate, wherein the embedded channel structure is aligned with the bottom channel structure. In accordance with some embodiments, the substrate may include an isolation structure laterally surrounding the bottom channel structure, and the sidewall channel structure is disposed on the isolation structure. In accordance with some embodiments, an inner spacer is further disposed between the embedded channel structure and the bottom channel structure. In accordance with some embodiments, a source/drain structure is further disposed on the substrate and in contact with the embedded channel structure and the sidewall channel structure.

(68) In accordance with some embodiments, a semiconductor device may include an embedded channel structure; a sidewall channel structure disposed at a lateral side of the embedded channel structure; an inner spacer disposed between the embedded channel structure and the sidewall channel structure; and a gate electrode structure surrounded by the inner spacer, the embedded channel structure and the sidewall channel structure. In accordance with some embodiments, a side surface of the inner spacer, a side surface of the sidewall channel structure and a side surface of the embedded channel structure are co-planar. In accordance with some embodiments, a gate dielectric layer further surrounds the gate electrode structure and the gate dielectric layer is surrounded by the inner spacer, the embedded channel structure and the sidewall channel structure. In accordance with some embodiments, a source/drain structure may be in contact with the embedded channel structure and the sidewall channel structure. In accordance with some embodiments, the inner spacer is disposed between the source/drain structure and the gate electrode structure. In accordance with some embodiments, a dielectric fin is further included, wherein the sidewall channel structure is disposed between the dielectric fin and the gate electrode structure. In accordance with some embodiments, a spacer is further disposed on the inner spacer and beside the gate electrode structure.

(69) In accordance with some embodiments, a method of fabricating a semiconductor device may include forming an epitaxial stack on a substrate; forming an epitaxial cap covering the epitaxial stack; patterning the epitaxial stack and the epitaxial cap to form an embedded channel structure and a sidewall channel structure located at a lateral side of the embedded channel structure; and forming a gate electrode structure surrounding the embedded channel structure and between the embedded channel structure and the sidewall channel structure. In accordance with some embodiments, the epitaxial cap covers a top surface and a side surface of the epitaxial stack. In accordance with some embodiments, the embedded channel structure and the sidewall channel structure have different crystal orientations. In accordance with some embodiments, a dielectric fin is further formed at a lateral side of the epitaxial cap, wherein the sidewall channel structure is in contact with the dielectric fin.

(70) It will be apparent to those skilled in the art that various modifications and variations can be made to the disclosed embodiments without departing from the scope or spirit of the disclosure. In view of the foregoing, it is intended that the disclosure covers modifications and variations provided that they fall within the scope of the following claims and their equivalents.

Claims

1. A semiconductor device, comprising: an embedded channel structure disposed on a substrate; a sidewall channel structure disposed on the substrate, and located at a lateral side of the embedded channel structure, wherein the embedded channel structure and the sidewall channel structure have different crystal orientations; and a gate electrode structure disposed on the substrate, encircling the

- embedded channel structure and located between the embedded channel structure and the sidewall channel structure.
2. The semiconductor device of claim 1, further comprising a dielectric fin disposed on the substrate, wherein the sidewall channel structure is in contact with a sidewall of the dielectric fin.
 3. The semiconductor device of claim 2, further comprising a gate dielectric layer disposed between the gate electrode structure and the sidewall channel structure, wherein the sidewall channel structure is disposed between the gate dielectric layer and the dielectric fin.
 4. The semiconductor device of claim 1, wherein the embedded channel structure is separated from the substrate by a distance.
 5. The semiconductor device of claim 1, further comprising a bottom channel structure formed in the substrate, wherein the embedded channel structure is aligned with the bottom channel structure.
 6. The semiconductor device of claim 5, wherein the substrate includes an isolation structure laterally surrounding the bottom channel structure, and the sidewall channel structure is disposed on the isolation structure.
 7. The semiconductor device of claim 5, further comprising an inner spacer disposed between the embedded channel structure and the bottom channel structure.
 8. The semiconductor device of claim 1, further comprising a source/drain structure disposed on the substrate and in contact with the embedded channel structure and the sidewall channel structure.
 9. A semiconductor device, comprising: a dielectric fin, wherein a gap is surrounded by a sidewall of the dielectric fin; an embedded channel structure disposed in the gap; a sidewall channel structure in contact with the sidewall of the dielectric fin, wherein the embedded channel structure and the sidewall channel structure have different crystal orientations; and a gate electrode structure disposed between the embedded channel structure and the sidewall channel structure.
 10. The semiconductor device of claim 9, further comprising a gate dielectric layer disposed between the gate electrode structure and the sidewall channel structure.
 11. The semiconductor device of claim 10, wherein the gate dielectric layer extends along the sidewall channel structure and the dielectric fin.
 12. The semiconductor device of claim 10, wherein the gate dielectric layer includes a dielectric portion disposed between the gate electrode structure and the embedded channel structure.
 13. The semiconductor device of claim 9, further comprising a source/drain structure in contact with the embedded channel structure and the sidewall channel structure.
 14. The semiconductor device of claim 9, wherein the dielectric fin has a higher height than the sidewall channel structure.
 15. The semiconductor device of claim 9, wherein the sidewall channel structure has an elongate shape extending in a Z-axis.
 16. A semiconductor device, comprising: a bottom channel structure; an isolation structure disposed beside the bottom channel structure; an embedded channel structure disposed over the bottom channel structure; a sidewall channel structure disposed on the isolation structure; and a gate electrode structure encircling the embedded channel structure and located between the embedded channel structure and the sidewall channel structure.
 17. The semiconductor device of claim 16, wherein the bottom channel structure is protruded relative to the isolation structure.
 18. The semiconductor device of claim 16, further comprising a gate dielectric layer disposed between the gate electrode structure and the sidewall channel structure, between the gate electrode structure and the isolation structure and the gate electrode structure and the bottom channel structure.
 19. The semiconductor device of claim 16, wherein the embedded channel structure and the sidewall channel structure have different crystal orientations.
 20. The semiconductor device of claim 16, wherein the sidewall channel structure has an elongate shape extending in a Z-axis.

